

SOFTWARE ENGINEERING LAB – 2

Agile Backlog Creation & Sprint Simulation in Jira

Smart Lab Equipment Slot Reservation Portal

Team: 001

Problem Statement: 001

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1. Project Overview

This document presents the Jira Scrum simulation for the Smart Lab Equipment Slot Reservation Portal. The Lab-2 activity converts the project requirements into Agile Epics and User Stories, prioritizes and estimates the backlog using story points, runs two simulated one-week sprints, and analyses sprint progress using a Burndown Chart.

Note: The sprint activity in this document represents the Jira-based Scrum simulation required for the laboratory exercise.

2. Epics

Paste the Jira screenshot showing the Epics here.

[PASTE SCREENSHOT – EPICS]

Epic	Epic Name
Epic 1	Equipment Availability Management
Epic 2	Equipment Reservation Management
Epic 3	Equipment Calibration Management
Epic 4	Equipment Return & Tracking

3. Agile Backlog – User Stories

Paste the Jira backlog/issues screenshot here.

[PASTE SCREENSHOT – BACKLOG / ALL USER STORIES]

User stories and estimates:

Issue Key	User Story	Priority	Story Points
SLESRPS-12	View Equipment Availability	High	3
SLESRPS-13	Search / Filter Equipment	High	2
SLESRPS-15	Reserve Equipment Slot	High	5
SLESRPS-16	Check Slot Availability	High	3
SLESRPS-17	Lock Slot & Generate Reservation Token	High	5
	Verify Calibration Status	High	3
	Update Equipment Availability	High	5
	Update Calibration Status	High	5
	Record Equipment Return	Medium	3
	Manage Return & Late-Return Rules	Medium	3
	TOTAL		37

4. Story Point Assignment

Story points were assigned using the Fibonacci-style scale to represent relative effort, complexity, and uncertainty.

	Work	Assignee	Reporter	Priority	Status	Resolution	Created	Updated	Due date	
☐	❖ SLESRPS-6 Equipment Availability Management	Unassigned	MJ Manjunatha H J	Medium	To Do	Unresolved	Oct 05, 2026, 10:26 AM	Oct 05, 2026, 10:26 AM	None	...
☐	❑ SLESRPS-12 View Equipment Availability	Unassigned	MJ Manjunatha H J	High	Done	Done	Oct 05, 2026, 10:49 AM	Oct 05, 2026, 5:25 PM	None	...
☐	❑ SLESRPS-13 Search / Filter Equipment	Unassigned	MJ Manjunatha H J	High	Done	Done	Oct 05, 2026, 4:48 PM	Oct 05, 2026, 5:25 PM	None	...
☐	❖ SLESRPS-7 Equipment Reservation Management	Unassigned	MJ Manjunatha H J	Medium	To Do	Unresolved	Oct 05, 2026, 10:28 AM	Oct 05, 2026, 10:29 AM	None	...
☐	❑ SLESRPS-15 Reserve Equipment Slot	Unassigned	MJ Manjunatha H J	High	Done	Done	Oct 05, 2026, 4:54 PM	Oct 05, 2026, 5:25 PM	None	...
☐	❑ SLESRPS-16 Check Slot Availability	Unassigned	MJ Manjunatha H J	High	Done	Done	Oct 05, 2026, 4:57 PM	Oct 05, 2026, 5:25 PM	None	...
☐	❑ SLESRPS-17 Lock Slot & Generate Reservation Token	Unassigned	MJ Manjunatha H J	High	Done	Done	Oct 05, 2026, 4:59 PM	Oct 05, 2026, 5:25 PM	None	...
☐	❖ SLESRPS-8 Equipment Calibration Management	Unassigned	MJ Manjunatha H J	Medium	To Do	Unresolved	Oct 05, 2026, 10:29 AM	Oct 05, 2026, 10:29 AM	None	...
☐	❑ SLESRPS-18 Verify Calibration Status	Unassigned	MJ Manjunatha H J	High	Done	Done	Oct 05, 2026, 5:00 PM	Oct 05, 2026, 5:31 PM	None	...
☐	❑ SLESRPS-19 Update Equipment Availability	Unassigned	MJ Manjunatha H J	High	Done	Done	Oct 05, 2026, 5:01 PM	Oct 05, 2026, 5:31 PM	None	...
☐	❑ SLESRPS-20 Update Calibration Status	Unassigned	MJ Manjunatha H J	High	Done	Done	Oct 05, 2026, 5:03 PM	Oct 05, 2026, 5:31 PM	None	...
☐	❖ SLESRPS-9 Equipment Return & Tracking	Unassigned	MJ Manjunatha H J	Medium	To Do	Unresolved	Oct 05, 2026, 10:29 AM	Oct 05, 2026, 10:30 AM	None	...
☐	❑ SLESRPS-21 Record Equipment Return	Unassigned	MJ Manjunatha H J	Medium	Done	Done	Oct 05, 2026, 5:05 PM	Oct 05, 2026, 5:31 PM	None	...
☐	❑ SLESRPS-22 Manage Return & Late-Return Rules	Unassigned	MJ Manjunatha H J	Medium	Done	Done	Oct 05, 2026, 5:07 PM	Oct 05, 2026, 5:31 PM	None	...

5. Sprint 1

Sprint duration: 1 week

Start date: 28 September 2026

End date: 5 October 2026

Sprint Goal: Enable students to find available laboratory equipment and successfully reserve an available time slot.

Sprint 1 planned stories:

User Story	Story Points
View Equipment Availability	3
Search / Filter Equipment	2
Reserve Equipment Slot	5
Check Slot Availability	3
Lock Slot & Generate Reservation Token	5
Sprint 1 Total	18

Report: SLESRPS Sprint 1 *Work item added after sprint start

Scope changes log [View in All work navigator](#)

Date ↑	Key	Summary	Work type	Epic	Details of scope change	Change in estimation
2026-10-05	SLESRPS-12	View Equipment Availability	Story	Equipment Availabi...	Estimate of 3 has been added	- → 3
2026-10-05	SLESRPS-13	Search / Filter Equipment	Story	Equipment Availabi...	Estimate of 2 has been added	- → 2
2026-10-05	SLESRPS-15	Reserve Equipment Slot	Story	Equipment Reserv...	Estimate of 5 has been added	- → 5
2026-10-05	SLESRPS-16	Check Slot Availability	Story	Equipment Reserv...	Estimate of 3 has been added	- → 3
2026-10-05	SLESRPS-17	Lock Slot & Generate Reservation Token	Story	Equipment Reserv...	Estimate of 5 has been added	- → 5
2026-10-05	SLESRPS-12*	View Equipment Availability	Story	Equipment Availabi...	Work item added to sprint	3
2026-10-05	SLESRPS-15*	Reserve Equipment Slot	Story	Equipment Reserv...	Work item added to sprint	5
2026-10-05	SLESRPS-16*	Check Slot Availability	Story	Equipment Reserv...	Work item added to sprint	3
2026-10-05	SLESRPS-17*	Lock Slot & Generate Reservation Token	Story	Equipment Reserv...	Work item added to sprint	5
2026-10-05	SLESRPS-13*	Search / Filter Equipment	Story	Equipment Availabi...	Work item added to sprint	2

Completed work items [View in All work navigator](#)

Key	Summary	Work type	Epic	Status	Assignee	Story points
SLESRPS-12	View Equipment Availability	Story	Equipment Availabi...	Done		3
SLESRPS-15	Reserve Equipment Slot	Story	Equipment Reserv...	Done		5
SLESRPS-16	Check Slot Availability	Story	Equipment Reserv...	Done		3
SLESRPS-17	Lock Slot & Generate Reservation Token	Story	Equipment Reserv...	Done		5
SLESRPS-13	Search / Filter Equipment	Story	Equipment Availabi...	Done		2

6. Sprint 2

Sprint duration: 1 week

Start date: 5 October 2026

End date: 12 October 2026

Sprint Goal: Ensure equipment calibration, technician updates, equipment returns, and late-return tracking are managed accurately.

Sprint 2 planned stories:

User Story	Story Points
Verify Calibration Status	3
Update Equipment Availability	5
Update Calibration Status	5
Record Equipment Return	3
Manage Return & Late-Return Rules	3
Sprint 2 Total	19

Report: SLESRPS Sprint 2 *Work item added after sprint start

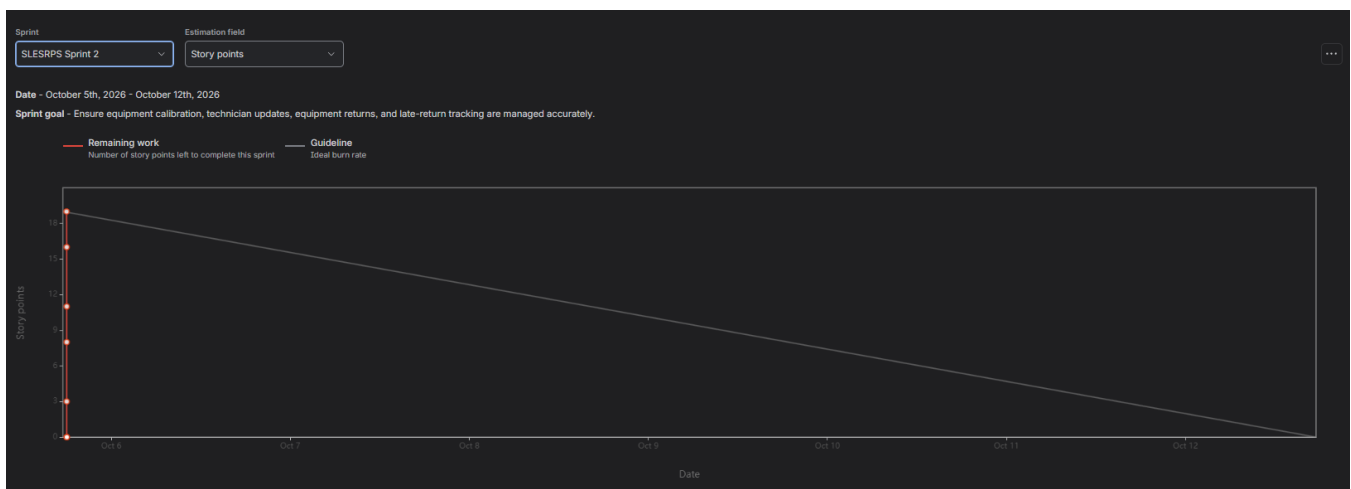
Scope changes log [View in All work navigator](#)

Date ↑	Key	Summary ↓	Work type	Epic	Details of scope change	Change in estimation
Sort descending There have been no scope changes this sprint						

Completed work items						View in All work navigator
Key	Summary	Work type	Epic	Status	Assignee	Story points
SLESRPS-18	Verify Calibration Status	Story	Equipment Calibrat...	Done		3
SLESRPS-19	Update Equipment Availability	Story	Equipment Calibrat...	Done		5
SLESRPS-20	Update Calibration Status	Story	Equipment Calibrat...	Done		5
SLESRPS-21	Record Equipment Return	Story	Equipment Return ...	Done		3
SLESRPS-22	Manage Return & Late-Return Rules	Story	Equipment Return ...	Done		3

7. Burndown Chart

The Burndown Chart was generated in Jira to observe the amount of remaining work during the simulated sprint.



Observation: The chart records the simulated Jira progress. The work was transitioned through the Scrum board during the lab session.

8. Reflection

1. Did your estimations reflect the actual effort?

The story point estimations reasonably represented the relative complexity of the requirements. Smaller stories such as Search / Filter Equipment were assigned 2 points, while more complex reservation and equipment-management stories were assigned 5 points. The estimates were intended to represent relative effort, complexity, and uncertainty rather than direct hours.

2. Was your backlog well-prioritized?

Yes. High-priority functionality related to equipment availability and reservation was prioritized first. This allowed Sprint 1 to focus on the main student availability and reservation flow, while Sprint 2 covered calibration, technician updates, equipment returns, and late-return management.

3. How did your simulated sprint align with your plan?

The simulated sprint structure aligned with the planned project flow. Sprint 1 focused on the student-facing availability and reservation activities, while Sprint 2 focused on calibration, technician management, and equipment return activities. The stories were moved through the Jira workflow to simulate sprint completion.

4. What insights did the burndown chart give about your team's capacity?

The Burndown Chart showed how the remaining estimated work changed as stories were completed. Because this was a simulated sprint performed in Jira during the lab session, the chart reflects the recorded simulation rather than actual development velocity. It demonstrates how a burndown can be used to monitor remaining work and compare progress with the sprint plan.

9. Final Summary

Item	Details
Project	Smart Lab Equipment Slot Reservation Portal
Team	001
Problem Statement	001
Epics	4
User Stories	10
Total Story Points	37
Sprint 1	18 story points
Sprint 2	19 story points
Sprints Simulated	2